Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]

[14EE46]

FOURTH SEMESTER B.TECH. EXAMINATION, JUNE – 2016
TRANSFORMERS AND INDUCTION MACHINES

Time: 3 Hours **Maximum Marks:70**

Instructions:

i. **Answer any five full questions. Choose Atleast two full questions from each part.**

ii. **Missing data may be suitably assumed.**

PART-A

1. a. Describe the operation of a single-phase transformer, **04**

   b. Explain clearly the functions of each part of a transformer . . **03**

   c. A 125KVA transformer having primary voltage of 2000V at 50Hz has 182 primary and 40 secondary turns. Neglecting losses calculate **07**

      i. Full load primary & secondary current.

      ii. No-load secondary induced e.m.f

      iii. The maximum flux in the curve.

2. a. Derive the condition for maximum efficiency. **05**

   b. Develop the exact equivalent circuit of a 1-ph transformer. Derive the approximate and simplified equivalent circuit of the transformer. **09**

3. a. With the help of relevant circuit diagram explain the back-to-back test on transformers. Mention the advantages. **08**

   b. Two single phase transformers share a load of 400KVA at power factor of 0.8 lag. Their equivalent impedance referred to secondary winding are $(1+j2.5)\Omega$ and $(1.5+j3)\Omega$ respectively. Calculate the load shared by each transformer. **06**

4. a. Draw the Scott connection of transformers and mark the terminals and turns ratio and explain. **06**

   b. A 400KVA load at 0.7 pf lagging is supplied by three single phase transformer connected in Δ-Δ. Each of Δ-Δ transformer is rated at 200KVA, 2300/230 V. If one defective transformer is removed from service, calculate for V-V connection **08**

      i. The KVA load carried by each transformer.

      ii. Percent rated load carried by each transformer.

      iii. Total KVA ratings of the transformer bank in V-V.

      iv. Ratio of V-V bank to Δ-Δ bank transformer rating.

PART-B

5. a. Explain the concept of rotating magnetic field in 3-phase induction motors. **06**

   b. A 746KW, 3-phase, 50HZ, 16-pole induction motor has a rotor impedance of $(0.02+j0.15)\Omega$ at stand still. Full load torque is obtained at 360rpm. Calculate. **08**

      i. The speed at which maximum torque occurs

      ii. The ratio of maximum to full-load torque

      iii. The external resistance/Phase to be inserted in the rotor circuit to get

maximum torque at starting.

6.	a.	Explain cogging and crawling phenomenon in 3–ph induction motor.	07
	b.	Describe with sketch, the construction of a double cage induction motor.	07
7.	a.	With neat sketch, explain auto-transformer starter.	06
	b.	Briefly explain different speed control methods of 3-ph induction motor.	08
8.	a.	With neat diagram explain shaded pole single phase motor.	06
	b.	Why a single phase induction motor is not self-starting. Explain the double revolving field theory.	08